

Global PC Component Price Intelligence Platform

Technical Specification, Epics and Delivery Backlog

Document status: Initial implementation specification

Target release: Private beta MVP

Initial delivery period: 12 weeks

Sprint length: Two weeks

Initial markets: United States, Germany/Eurozone, United Kingdom and Japan

Initial categories: DDR4 memory, DDR5 memory and consumer NVMe SSDs

1. Purpose

This document converts the product plan into a technical implementation specification.

It defines:

- System architecture
- Core services
- Data model
- Price-ingestion workflow
- Product-matching workflow
- Index calculations
- Editorial and forecasting workflows
- Public API requirements
- User-facing features
- Administration features
- Security and privacy requirements
- Engineering epics
- User stories
- Acceptance criteria
- Sprint sequencing
- Release gates

The MVP must allow the team to collect global and regional component-price data, publish market events and analysis, calculate regional price indices and publish accountable buy-or-wait forecasts.

2. MVP scope

2.1 Included

The MVP will support:

- Global semiconductor-market overview
- Four retail markets
- Regional currencies
- Regional retailer data
- Canonical component catalogue
- Daily price collection
- Immutable raw price history
- Daily category indices
- Regional price comparison
- Global market events
- Regional event impacts
- Editorial analysis
- Buy-or-wait recommendations
- Forecast publication
- Forecast outcome tracking
- Public methodology
- Affiliate-link tracking
- Newsletter signup
- Internal editorial and data-administration interface
- Basic operational monitoring
- Mobile-responsive public website

2.2 Excluded

The MVP will not include:

- User accounts
- Saved watchlists
- Personalised alerts
- Paid subscriptions
- Public API access for third parties
- Automated AI-written articles
- Automated forecasts without human approval
- GPU or CPU tracking
- Full build-price calculation
- Native mobile applications
- Live intraday pricing
- Full multilingual publication
- Retailer bidding or sponsored product rankings
- Complete worldwide retail coverage

3. Product principles

3.1 Raw data must remain immutable

Imported price observations must never be overwritten.

Corrections must create separate correction records or updated derived records while preserving the original observation.

3.2 Facts and analysis must be separated

The system must distinguish:

- Source facts
- Editorial interpretation
- Forecast assumptions
- Consumer recommendation

3.3 All public values must be traceable

Every displayed price, index, event and forecast must be traceable to:

- Source
- Collection date
- Calculation version
- Editorial author or automated process
- Last review date

3.4 Incomplete data must be visible

The system must not conceal weak market coverage.

Every regional index must display a data-quality state.

3.5 Commercial relationships must not affect analysis

Affiliate commission, sponsorship or retailer relationships must not alter:

- Product eligibility
- Index composition
- Event impact
- Forecast direction
- Consumer recommendation

4. Recommended system architecture

The implementation should use a modular monolith for the MVP.

A microservice architecture would add unnecessary operational complexity at this stage.

The system should be divided logically into modules that can later be separated if required.

4.1 Core components

Public web application

Responsibilities:

- Homepage
- Global market pages
- Regional pages
- Category pages
- Product pages
- Event pages
- Forecast pages
- Methodology
- Affiliate-link redirection
- Newsletter signup

Internal administration application

Responsibilities:

- Product review
- Listing matching
- Retailer administration
- Region administration
- Event creation
- Source management
- Forecast creation
- Forecast review
- Correction management
- Data-quality review
- Import monitoring

The public and administration applications may initially share the same codebase with role-based route protection.

Application API

Responsibilities:

- Public read endpoints
- Administration endpoints
- Data queries
- Validation
- Authentication
- Index retrieval
- Forecast retrieval
- Event and source relationships
- Affiliate redirects

Relational database

Responsibilities:

- Canonical entities
- Product catalogue
- Raw observations
- Derived daily prices
- Category baskets
- Indices
- Events
- Sources
- Forecasts
- Corrections
- Operational records

PostgreSQL is recommended.

Scheduled job runner

Responsibilities:

- Retailer imports
- Affiliate-feed imports
- Marketplace API imports
- Exchange-rate imports
- Daily price derivation
- Daily index generation
- Forecast review reminders
- Data-quality checks
- Stale-data detection

Object storage

Responsibilities:

- Raw feed files
- Import snapshots
- Source documents where permitted
- Generated report assets
- Chart exports
- Audit artefacts

Monitoring and alerting

Responsibilities:

- Import failures
 - API failures
 - Missing daily data
 - Stale exchange rates
 - Large price anomalies
 - Database errors
 - Background-job failures
 - Public-site availability
-

5. Suggested technology stack

The exact implementation language may be changed according to team skills.

5.1 Reference stack

Front end

- Next.js
- TypeScript
- Server-rendered public pages
- Responsive web design
- Accessible charting library
- Static generation where practical

Back end

- Next.js server routes or a separate TypeScript application
- Node.js
- Type-safe validation
- Background worker process

Database

- PostgreSQL
- Database migrations committed to source control
- SQL or a mature ORM
- Read replicas only when scale justifies them

Job scheduling

One of:

- Scheduled server jobs
- GitHub Actions
- Cloud scheduler
- Queue-based worker
- Container cron process

Production jobs must use locking to prevent accidental duplicate execution.

Authentication

- Passwordless or managed authentication for internal users
- Role-based access control
- Multi-factor authentication for administrative accounts where available

Charts

The chosen chart library must support:

- Time-series lines
- Event annotations
- Tooltips
- Region comparison
- Currency formatting
- Mobile layouts
- Accessible labels
- Data export later

6. Environments

The project must have at least three environments.

6.1 Local development

Used for:

- Engineering development
- Test imports
- Local database migrations
- Unit tests

Production credentials must never be used locally.

6.2 Staging

Used for:

- Team review
- Private beta testing
- Data-source integration testing
- Editorial workflow testing
- Release candidate validation

Staging should use separate retailer and affiliate credentials where providers support them.

6.3 Production

Used for:

- Live data
 - Public pages
 - Real affiliate links
 - Final editorial publication
 - Production monitoring
-

7. User roles and permissions

7.1 Public visitor

Can:

- View public pages
- Select region
- View current and historical prices
- Read event analysis
- Read forecasts

- Use affiliate links
- Subscribe to the newsletter

Cannot:

- Modify data
- View unpublished editorial content
- View internal data-quality queues

7.2 Data reviewer

Can:

- Review unmatched listings
- Approve or reject product matches
- Correct retailer metadata
- Mark invalid observations
- Review anomalies

Cannot:

- Publish editorial analysis
- Publish forecasts
- Change user permissions

7.3 Editor

Can:

- Create and edit events
- Manage sources
- Create regional impact assessments
- Draft forecasts
- Publish articles
- Submit corrections

Cannot:

- Change raw price observations
- Manage system users unless separately authorised

7.4 Senior editor

Can:

- Publish forecasts
- Review forecast outcomes

- Publish corrections
- Override recommendations with documented justification
- Approve methodology changes

7.5 Administrator

Can:

- Manage users
- Manage regions
- Manage retailers
- Configure data sources
- View system logs
- Retry jobs
- Manage feature flags

7.6 Commercial user

Can:

- Configure affiliate identifiers
- View click analytics
- Manage sponsor disclosure metadata
- Review commercial reports

Cannot:

- Change index eligibility
- Change forecasts
- Change editorial conclusions

8. Core data model

All main records should use UUID primary keys unless there is a strong technical reason not to.

All records should include:

- `created_at`
- `updated_at`
- `created_by` where applicable
- `updated_by` where applicable

Timestamps should be stored in UTC.

9. Geographic and currency entities

9.1 Region

Represents a consumer-facing geographic market.

Suggested fields:

- id
- code
- name
- slug
- region_group
- country_code
- default_currency_id
- tax_display_type
- is_public
- is_retail_supported
- coverage_status
- launch_date
- methodology_notes

Example values:

- United States
- Germany
- United Kingdom
- Japan
- Eurozone overview

9.2 Currency

Fields:

- id
- iso_code
- name
- symbol
- decimal_places
- display_format

9.3 Exchange rate

Fields:

- id
- base_currency_id
- quote_currency_id
- rate
- rate_date
- provider
- source_reference
- collected_at

Constraints:

- Unique base, quote and date
 - No future-dated rates unless explicitly marked as projections
-

10. Retail and affiliate entities

10.1 Retailer

Fields:

- id
- name
- slug
- website_domain
- region_id
- retailer_type
- tax_included_by_default
- marketplace_allowed
- is_active
- data_quality_rating
- affiliate_program_id
- notes

10.2 Affiliate programme

Fields:

- id
- name
- network

- programme_reference
- default_commission_rate
- cookie_duration_days
- terms_url
- permitted_data_usage
- historical_storage_allowed
- is_active
- reviewed_at

10.3 Affiliate link template

Fields:

- id
- affiliate_program_id
- region_id
- url_template
- tracking_parameter_name
- campaign_identifier
- sub_id_supported
- is_active

10.4 Affiliate click

Fields:

- id
- occurred_at
- region_id
- retailer_id
- canonical_product_id
- source_page_type
- source_page_id
- campaign_code
- anonymous_session_reference
- destination_url
- consent_state

Do not store unnecessary personally identifiable information.

11. Product catalogue entities

11.1 Manufacturer

Fields:

- id
- name
- slug
- website
- country_code
- is_active

11.2 Product category

Fields:

- id
- name
- slug
- parent_category_id
- unit_type
- is_public
- display_order

Initial categories:

- Desktop memory
- DDR4 desktop memory
- DDR5 desktop memory
- NVMe SSD
- PCIe 4.0 NVMe SSD

11.3 Canonical product

Represents one exact manufacturer product.

Fields:

- id
- manufacturer_id
- category_id
- manufacturer_part_number
- gtin
- upc

- ean
- name
- slug
- status
- release_date
- discontinued_date
- successor_product_id
- technical_specification_json
- primary_image_url
- is_index_eligible
- review_status

Relevant memory specifications:

- Memory generation
- Total capacity
- Module count
- Capacity per module
- Speed
- CAS latency
- Voltage
- ECC
- Registered or unbuffered
- Desktop DIMM or laptop SODIMM
- XMP support
- EXPO support

Relevant SSD specifications:

- Capacity
- Interface
- PCIe generation
- Form factor
- NAND type where known
- DRAM cache
- Rated read speed
- Rated write speed
- Endurance
- Heatsink included

11.4 Retailer listing

Represents a retailer-specific listing.

Fields:

- id

- retailer_id
- external_listing_id
- canonical_product_id
- listing_title
- listing_url
- retailer_sku
- manufacturer_part_number_raw
- gtin_raw
- matching_status
- match_confidence
- match_method
- first_seen_at
- last_seen_at
- is_active
- is_marketplace_listing
- seller_name
- listing_metadata_json

Matching statuses:

- Unmatched
- Suggested
- Confirmed
- Rejected
- Ambiguous
- Retired

12. Price entities

12.1 Raw price observation

This is an immutable record.

Fields:

- id
- retailer_listing_id
- observed_at
- currency_id
- item_price
- shipping_price
- tax_amount
- total_display_price
- tax_included

- stock_status
- stock_quantity
- seller_name
- promotion_text
- source_type
- source_import_id
- source_payload_reference
- is_valid
- invalid_reason

Stock statuses:

- In stock
- Low stock
- Backorder
- Preorder
- Out of stock
- Unknown

12.2 Daily product price

Derived from raw observations.

Fields:

- id
- canonical_product_id
- region_id
- price_date
- currency_id
- minimum_price
- median_price
- maximum_price
- qualifying_offer_count
- retailer_count
- in_stock_offer_count
- calculation_version
- quality_status
- generated_at

12.3 Price correction

Fields:

- id
- raw_price_observation_id

- `correction_type`
- `reason`
- `replacement_value_json`
- `approved_by`
- `approved_at`

Corrections must not delete the original record.

13. Index entities

13.1 Index definition

Represents one tracked basket.

Fields:

- `id`
- `name`
- `slug`
- `region_id`
- `category_id`
- `currency_id`
- `description`
- `calculation_method`
- `outlier_method`
- `minimum_product_count`
- `minimum_retailer_count`
- `tax_basis`
- `is_public`
- `effective_from`
- `effective_to`
- `calculation_version`

Examples:

- US 32GB DDR5-6000 Index
- UK 64GB DDR5-6000 Index
- Germany 2TB PCIe 4.0 SSD Index

13.2 Index eligibility rule

Fields:

- `id`

- index_definition_id
- rule_type
- field_name
- operator
- rule_value
- display_description

Example rules:

- Memory generation equals DDR5
- Total capacity equals 32GB
- Speed between 5600 and 6400
- Module count equals two
- ECC equals false

13.3 Index membership

Records the products included for a defined period.

Fields:

- id
- index_definition_id
- canonical_product_id
- effective_from
- effective_to
- inclusion_reason
- approved_by

13.4 Daily index value

Fields:

- id
- index_definition_id
- index_date
- raw_median_price
- indexed_value
- minimum_price
- maximum_price
- product_count
- retailer_count
- coverage_percentage
- quality_status
- calculation_version
- generated_at

13.5 Index composition change

Fields:

- id
 - index_definition_id
 - change_date
 - change_type
 - product_id
 - reason
 - approved_by
 - notes
-

14. Event and source entities

14.1 Source organisation

Fields:

- id
- name
- source_type
- website
- reliability_tier
- notes

Source types:

- Manufacturer
- Regulator
- Government
- Market research
- Financial publication
- Technology publication
- Retailer
- Community
- Other

14.2 Source item

Represents an individual article, announcement or report.

Fields:

- id
- source_organisation_id
- title
- url
- publication_date
- accessed_at
- author
- content_type
- rights_notes
- summary
- is_primary_source

14.3 Market event

Fields:

- id
- title
- slug
- event_date
- publication_status
- global_summary
- fact_summary
- editorial_analysis
- event_type
- default_direction
- default_strength
- expected_delay_min_days
- expected_delay_max_days
- confidence
- review_date
- author_id
- published_at

14.4 Event source relationship

Fields:

- id
- market_event_id
- source_item_id
- relationship_type
- supports_event
- supports_interpretation

- `contradicts_interpretation`
- `editor_notes`

14.5 Event category impact

Fields:

- `id`
- `market_event_id`
- `category_id`
- `impact_direction`
- `impact_strength`
- `expected_delay_min_days`
- `expected_delay_max_days`
- `confidence`
- `rationale`

14.6 Event regional impact

Fields:

- `id`
- `market_event_id`
- `region_id`
- `impact_direction`
- `impact_strength`
- `currency_effect`
- `inventory_effect`
- `expected_delay_min_days`
- `expected_delay_max_days`
- `confidence`
- `rationale`
- `reviewed_at`

15. Forecast entities

15.1 Forecast

Fields:

- `id`
- `title`
- `slug`
- `region_id`

- index_definition_id
- published_at
- forecast_start_date
- forecast_end_date
- direction
- expected_change_min_percentage
- expected_change_max_percentage
- confidence_percentage
- consumer_recommendation
- summary
- supporting_evidence
- contradictory_evidence
- invalidation_conditions
- status
- author_id
- approved_by

Forecast statuses:

- Draft
- Pending approval
- Published
- Under review
- Scored
- Withdrawn

15.2 Forecast evidence link

Fields:

- id
- forecast_id
- market_event_id
- source_item_id
- relationship_type
- weight
- notes

15.3 Forecast result

Fields:

- id
- forecast_id
- reviewed_at
- starting_index_value

- ending_index_value
- actual_change_percentage
- actual_direction
- direction_correct
- within_forecast_range
- midpoint_error_percentage_points
- result_classification
- review_notes
- reviewed_by

Result classifications:

- Correct
- Mostly correct
- Partially correct
- Incorrect
- Unable to score

16. Operational entities

16.1 Data source configuration

Fields:

- id
- name
- source_type
- region_id
- retailer_id
- schedule
- configuration_json
- credential_reference
- is_active
- last_success_at
- last_failure_at

Credentials must not be stored directly in normal database fields.

16.2 Import run

Fields:

- id
- data_source_configuration_id

- started_at
- completed_at
- status
- records_received
- records_created
- records_updated
- records_rejected
- warning_count
- error_message
- raw_file_reference

16.3 Data-quality issue

Fields:

- id
- issue_type
- severity
- entity_type
- entity_id
- description
- detected_at
- status
- assigned_to
- resolved_at
- resolution_notes

17. Data-ingestion workflow

Each source integration should follow the same pipeline.

17.1 Import stages

1. Fetch data.
2. Save the raw response or feed file.
3. Validate the source format.
4. Parse records.
5. Identify or create retailer listings.
6. Attempt canonical product matching.
7. Create immutable raw price observations.
8. Run anomaly checks.
9. Mark records as valid or requiring review.
10. Produce an import summary.
11. Trigger downstream daily derivation where appropriate.

17.2 Import idempotency

Repeated processing of the same source response must not produce duplicate records.

Suggested uniqueness:

- Source
- External listing ID
- Observation timestamp
- Price
- Source record identifier

17.3 Failure handling

When an import fails:

- Record the failure.
 - Preserve partial progress only where safe.
 - Alert the responsible team.
 - Do not publish incomplete derived data silently.
 - Allow an authorised user to retry the import.
 - Prevent simultaneous retries.
-

18. Product-matching workflow

18.1 Matching priority

The automated matcher should use:

1. Exact manufacturer part number
2. Exact GTIN, UPC or EAN
3. Existing retailer SKU mapping
4. Normalised manufacturer and part-number combination
5. Structured specification match
6. Fuzzy title match as a last resort

18.2 Match confidence

Suggested thresholds:

- 0.98–1.00: automatic confirmation
- 0.90–0.979: suggested confirmation
- 0.70–0.899: manual review required
- Below 0.70: unmatched

Automatic thresholds should be configurable.

18.3 Manual review screen

The reviewer must see:

- Retailer listing title
- Retailer URL
- Extracted manufacturer
- Extracted part number
- Extracted specifications
- Suggested canonical products
- Differences between listing and canonical product
- Historical reviewer decisions
- Confidence score

Available actions:

- Confirm match
- Select different product
- Create new canonical product
- Reject listing
- Mark as ambiguous
- Mark as unsupported category

19. Price qualification rules

A raw observation should be excluded from normal indices when:

- Product matching is unconfirmed
- Listing is out of stock
- Listing is a marketplace seller and the index excludes marketplaces
- Product condition is used or refurbished
- Listing is a bundle
- Shipping cost is unavailable where materially relevant
- Price is clearly erroneous
- Retailer listing has changed product
- Tax treatment is unknown and cannot be normalised
- Listing violates regional methodology

Excluded observations remain stored.

20. Daily product-price calculation

For each canonical product and region:

1. Select valid observations within the relevant collection window.
2. Use the latest qualifying observation per retailer.
3. Calculate total consumer price according to regional methodology.
4. Calculate minimum, median and maximum.
5. Count offers and retailers.
6. Assign quality state.
7. Store the daily derived record.

Suggested collection window:

- Latest valid observation within 36 hours for a daily index

If insufficient data exists:

- Do not fabricate a value.
- Mark the day as incomplete.
- Use gaps in the chart or a clearly labelled carry-forward rule if the methodology permits it.

21. Daily index calculation

21.1 MVP calculation

For each index:

1. Retrieve eligible products with valid daily product prices.
2. Enforce minimum retailer and product counts.
3. Apply documented outlier rules.
4. Calculate the median regional price.
5. Calculate index value relative to the defined baseline.
6. Store coverage and quality metadata.
7. Save the calculation version.

21.2 Indexed comparison

Formula:

$$\text{Indexed Value} = \text{Current Basket Value} / \text{Baseline Basket Value} \times 100$$

The baseline date must be visible.

When index composition changes materially:

- Either chain the index using a documented method, or
- Begin a new calculation version.

21.3 Data-quality states

High confidence

- Meets or exceeds target product count
- Meets retailer target
- Majority of products in stock
- No unresolved severe anomalies

Moderate confidence

- Minimum publication threshold met
- Some missing products or retailers
- No severe integrity issue

Limited confidence

- Barely meets minimum threshold
- Low stock coverage
- Significant regional gaps

Insufficient data

- Publication threshold not met
 - Index value not published as authoritative
-

22. Public API specification

The MVP API is internal to the public site but should be structured cleanly for future external use.

22.1 General requirements

- JSON responses
- Versioned endpoints
- Stable identifiers
- Pagination
- Input validation
- Rate limiting
- Caching for public data
- No unpublished content returned publicly

- Consistent error format

22.2 Proposed public endpoints

Regions

```
GET /api/v1/regions
```

Returns supported regions.

```
GET /api/v1/regions/{slug}
```

Returns regional metadata.

Categories

```
GET /api/v1/categories
```

```
GET /api/v1/categories/{slug}
```

Indices

```
GET /api/v1/indices
```

Filters:

- Region
- Category
- Status

```
GET /api/v1/indices/{slug}
```

```
GET /api/v1/indices/{slug}/history
```

Parameters:

- Start date
- End date
- Currency mode
- Indexed or raw

Products

```
GET /api/v1/products/{slug}
```

```
GET /api/v1/products/{slug}/prices
```

Events

GET /api/v1/events

Filters:

- Region
- Category
- Event type
- Date range

GET /api/v1/events/{slug}

Forecasts

GET /api/v1/forecasts

GET /api/v1/forecasts/{slug}

GET /api/v1/forecast-scorecard

Market overview

GET /api/v1/market-overview

Returns:

- Global status
- Regional summaries
- Featured indices
- Recent events
- Active recommendations
- Scorecard summary

23. Administration API

Administration endpoints must require authentication and appropriate permissions.

Examples:

- Create and update canonical products
- Confirm listing matches
- Retry imports
- Create events
- Attach sources

- Create regional impacts
- Draft forecasts
- Approve forecasts
- Score forecasts
- Publish corrections
- Update methodology version
- Manage retailers and regions

All administrative writes must create audit records.

24. Public page requirements

24.1 Homepage

Must display:

- Global DRAM status
- Global NAND status
- Current overall direction
- Active global forecast
- Regional comparison
- Featured price indices
- Recent market events
- Buy-or-wait summaries
- Forecast scorecard
- Newsletter signup

Acceptance criteria:

- A first-time visitor can identify the global direction within five seconds.
- Every status includes a last-updated time.
- Every recommendation includes confidence.
- Regional comparison works on mobile.
- No price is shown without currency and tax context.

24.2 Regional page

Must display:

- Region name
- Local currency
- Tax treatment
- Regional market status
- Regional indices
- Regional event impacts

- Currency effect
- Buy-or-wait recommendations
- Retailer links

24.3 Category page

Must display:

- Category description
- Global trend
- Regional trends
- Historical chart
- Event annotations
- Active forecast
- Forecast history
- Current products
- Methodology link

24.4 Product page

Must display:

- Exact canonical product
- Key specifications
- Regional retailer offers
- Price history
- Comparison with category median
- Current availability
- Affiliate disclosure
- Data freshness
- Error-report option

24.5 Event page

Must display:

- Event date
- Fact summary
- Editorial analysis
- Sources
- Categories affected
- Regional impacts
- Expected delay
- Confidence
- Related forecasts
- Review status

24.6 Forecast page

Must display:

- Region
 - Category
 - Forecast horizon
 - Expected direction
 - Expected range
 - Confidence
 - Recommendation
 - Supporting evidence
 - Contradictory evidence
 - Invalidation conditions
 - Outcome when available
-

25. Editorial administration requirements

25.1 Event editor

The editor must be able to:

- Enter title
- Enter event and publication dates
- Attach sources
- Mark primary sources
- Write fact summary
- Write analysis
- Select categories
- Select global and regional impacts
- Add expected delays
- Add confidence
- Set review date
- Preview and publish

25.2 Forecast editor

The editor must be able to:

- Select region and index
- Select time horizon
- Enter expected range
- Enter confidence
- Select recommendation

- Attach evidence
- Add contradictory evidence
- Add invalidation conditions
- Preview
- Submit for approval
- Publish

A forecast cannot be published without:

- At least one source or market event
- Supporting evidence
- Contradictory evidence or explicit statement that none was identified
- Review date
- Confidence level
- Approval by an authorised role

26. Forecast-scoring requirements

A scheduled job should detect forecasts whose end date has passed.

The system should:

1. Retrieve start and end index values.
2. Calculate actual change.
3. Determine actual direction.
4. Test whether direction was correct.
5. Test whether result fell within the forecast range.
6. Calculate midpoint error.
7. Create a draft forecast result.
8. Notify an editor.
9. Require editorial review before publication.

The public scorecard must not exclude failed forecasts unless they are marked unable to score with a visible reason.

27. Non-functional requirements

27.1 Performance

Targets:

- Public page initial response under one second where cached
- Main content usable within 2.5 seconds on a typical mobile connection

- API cached response under 500 milliseconds
- Administration searches under two seconds for normal datasets

27.2 Availability

MVP target:

- 99.5% monthly availability for the public site
- Scheduled import failures must not take down the public site

27.3 Accessibility

Target WCAG 2.2 AA.

Requirements:

- Keyboard navigation
- Visible focus states
- Semantic headings
- Accessible form labels
- Colour not used as the only meaning
- Text alternatives for chart conclusions
- Sufficient contrast
- Screen-reader-readable price and forecast values

27.4 Responsive design

Must support:

- 320px mobile width
- Tablets
- Standard desktop
- Wide desktop

Charts must not require horizontal scrolling for normal use.

27.5 Browser support

Support current stable versions of:

- Chrome
- Edge
- Firefox
- Safari
- Mobile Chrome
- Mobile Safari

27.6 Localisation readiness

Even if the MVP is English-only:

- Text must not be hard-coded throughout components.
 - Dates and currencies must use locale-aware formatting.
 - Region metadata must support local names.
 - Decimal and thousands separators must be configurable.
-

28. Security requirements

28.1 Administrative access

- Authentication required
- Strong password or passwordless authentication
- MFA where available
- Session expiration
- Role-based access
- Login attempt monitoring

28.2 Secrets

API keys and affiliate credentials must:

- Use a secret-management facility
- Never appear in source control
- Never be returned to the browser
- Be rotated following suspected exposure

28.3 Input security

- Validate all inputs
- Sanitize rendered editorial content
- Prevent script injection
- Restrict uploaded file types
- Use parameterised database queries
- Add rate limiting to public forms

28.4 Audit logging

Audit events should include:

- User login
- Permission changes

- Forecast publication
 - Event publication
 - Product-match override
 - Price correction
 - Index methodology change
 - Affiliate-configuration change
-

29. Privacy and consent requirements

The MVP should minimise personal data.

Potential personal data:

- Administrative user details
- Newsletter signup details
- Anonymous analytics identifiers
- Affiliate-click identifiers
- Contact-form submissions

Requirements:

- Clear privacy notice
 - Consent management for non-essential tracking
 - No advertising or affiliate tracker before consent where legally required
 - Data-retention schedule
 - Newsletter consent evidence
 - Unsubscribe support
 - User-data deletion process
 - Separation of necessary operational logs from marketing analytics
-

30. Analytics requirements

Track:

- Page views
- Unique sessions
- Region selection
- Chart date-range changes
- Category selection
- Forecast views
- Methodology views
- Affiliate clicks
- Newsletter signup

- Data-error reports

Do not collect unnecessary detailed behavioural data during the MVP.

31. SEO requirements

Every public entity should have:

- Unique title
- Unique meta description
- Canonical URL
- Open Graph metadata
- Structured heading hierarchy
- Server-rendered content
- Crawlable text summary of charts
- Sitemap inclusion
- Last-modified date

Structured data may be added where genuinely applicable, but must not misrepresent financial or product claims.

32. Testing strategy

32.1 Unit tests

Required for:

- Currency conversion
- Tax normalisation
- Product matching helpers
- Median calculations
- Index calculations
- Outlier filtering
- Forecast scoring
- Affiliate-link construction

32.2 Integration tests

Required for:

- Data-source imports
- Raw observation creation
- Product matching

- Daily derivation
- Daily index generation
- Event publication
- Forecast publication
- Forecast scoring

32.3 End-to-end tests

Core journeys:

1. Visitor selects a region.
2. Visitor views an index.
3. Visitor views an event annotation.
4. Visitor reads a forecast.
5. Visitor follows an affiliate link.
6. Editor creates and publishes an event.
7. Data reviewer resolves a product match.
8. Senior editor publishes a forecast.
9. Administrator retries a failed import.

32.4 Data regression tests

Maintain fixed sample datasets for:

- DDR4 memory
- DDR5 memory
- SSD listings
- Tax-inclusive prices
- Tax-exclusive prices
- Marketplace listings
- Discounted products
- Invalid bundles

Index calculations must produce repeatable results from fixed data.

33. Engineering epics

Epic E01: Project foundation

Goal

Create the development foundation, environments, deployment workflow and shared engineering standards.

Stories

E01-S01: Create source repositories

As a developer, I need version-controlled repositories so the team can collaborate safely.

Acceptance criteria:

- Repository structure agreed
- Branch protection enabled
- Pull-request review required
- Secret scanning enabled
- Code ownership configured
- README includes local setup

E01-S02: Configure local development

Acceptance criteria:

- Application runs locally
- Local database starts with one command
- Seed data available
- Environment-variable example supplied
- No production credentials required

E01-S03: Configure CI

Acceptance criteria:

- Linting runs on pull requests
- Type checking runs
- Unit tests run
- Build verification runs
- Failed checks block merge

E01-S04: Configure staging deployment

Acceptance criteria:

- Staging deploys automatically from the agreed branch
 - Database migrations run safely
 - Staging uses separate credentials
 - Rollback process documented
-

Epic E02: Geographic and retailer foundation

Goal

Create regional, currency, retailer and affiliate configuration.

Stories

E02-S01: Manage currencies

Acceptance criteria:

- Administrator can create and edit currencies
- Currency codes are unique
- Display formatting is locale-aware
- Unsupported decimal configurations are rejected

E02-S02: Manage regions

Acceptance criteria:

- Administrator can create regions
- Each region has a default currency
- Tax-display method is required
- Region can be hidden without deletion

E02-S03: Manage retailers

Acceptance criteria:

- Retailer belongs to a region
- Tax and marketplace settings are configurable
- Retailer can be disabled
- Affiliate programme can be attached

E02-S04: Import exchange rates

Acceptance criteria:

- Daily exchange rates are stored
- Duplicate daily rates are prevented
- Stale data creates an alert
- Conversion calculation is unit tested

Epic E03: Canonical product catalogue

Goal

Create a reliable canonical product catalogue for memory and SSDs.

Stories

E03-S01: Create manufacturers

E03-S02: Create product categories

E03-S03: Create canonical products

Acceptance criteria:

- Part number is required where available
- Specifications are validated by category
- Duplicate part numbers are flagged
- Product status is supported
- Products can be marked index eligible

E03-S04: Import initial product catalogue

Acceptance criteria:

- Initial 30–60 products created
- Specifications reviewed
- Products assigned to launch categories
- Every product has a canonical name and part number

Epic E04: Retailer data ingestion

Goal

Import retailer and marketplace data reliably.

Stories

E04-S01: Create generic import framework

Acceptance criteria:

- Import sources use a shared interface

- Raw source file is retained
- Import status is recorded
- Imports are retryable
- Duplicate imports are prevented

E04-S02: Implement first retailer source

E04-S03: Implement second retailer source

E04-S04: Implement marketplace source

E04-S05: Add scheduled imports

Acceptance criteria:

- Schedule is configurable
 - Concurrent duplicate runs are blocked
 - Failure alert is generated
 - Successful run summary is available
-

Epic E05: Product matching and review

Goal

Match retailer listings to canonical products accurately.

Stories

E05-S01: Exact part-number matching

E05-S02: Specification extraction

E05-S03: Confidence scoring

E05-S04: Manual review queue

Acceptance criteria:

- Reviewer sees suggested matches
- Reviewer can confirm, reject or create product
- Decision is audited
- Similar future listings benefit from approved mappings

E05-S05: Listing lifecycle management

Acceptance criteria:

- First and last seen dates recorded
 - Missing listings become inactive after configured period
 - Product changes are detected where possible
-

Epic E06: Price history and daily derivation

Goal

Store raw prices and calculate daily product prices.

Stories

E06-S01: Store immutable raw observations

E06-S02: Validate price observations

E06-S03: Qualify offers

E06-S04: Calculate daily product prices

E06-S05: Implement price corrections

Acceptance criteria:

- Original observation remains unchanged
 - Correction reason is required
 - Public calculations use approved correction
 - Correction appears in audit log
-

Epic E07: Category indices

Goal

Create transparent regional price indices.

Stories

E07-S01: Create index definitions

E07-S02: Configure eligibility rules

E07-S03: Manage index membership

E07-S04: Calculate daily index values

E07-S05: Calculate data-quality state

E07-S06: Display index history

Acceptance criteria:

- Raw and indexed modes supported
 - Baseline displayed
 - Missing days handled transparently
 - Quality state shown
 - Currency clearly displayed
-

Epic E08: Market events and sources

Goal

Allow editors to publish evidence-backed market events.

Stories

E08-S01: Manage source organisations

E08-S02: Manage source items

E08-S03: Create market events

E08-S04: Attach sources to events

E08-S05: Add category impact

E08-S06: Add regional impact

E08-S07: Publish event pages

Acceptance criteria:

- Facts and analysis are separate
 - Sources are visible
 - Event date and publication date are distinct
 - Confidence is required
 - Regional impacts can differ
-

Epic E09: Forecasting and recommendations

Goal

Publish transparent forecasts and track results.

Stories

E09-S01: Create forecast drafts

E09-S02: Add evidence and counterevidence

E09-S03: Approval workflow

E09-S04: Publish forecast pages

E09-S05: Detect forecasts requiring review

E09-S06: Calculate forecast outcomes

E09-S07: Publish forecast scorecard

Acceptance criteria:

- Failed forecasts remain visible
 - Confidence shown
 - Expected range shown
 - Recommendation shown
 - Outcome linked to original forecast
-

Epic E10: Public website

Goal

Deliver the public consumer experience.

Stories

E10-S01: Global homepage

E10-S02: Regional selector

E10-S03: Regional market page

E10-S04: Category page

E10-S05: Product page

E10-S06: Event page

E10-S07: Forecast page

E10-S08: Methodology page

E10-S09: Responsive navigation

E10-S10: Accessibility pass

Epic E11: Affiliate and commercial foundation

Goal

Support disclosed affiliate links and basic revenue measurement.

Stories

E11-S01: Configure affiliate programmes

E11-S02: Generate regional affiliate links

E11-S03: Create affiliate redirect endpoint

E11-S04: Record privacy-conscious click analytics

E11-S05: Display affiliate disclosure

Acceptance criteria:

- Disclosure is visible before commercial links
- Destination URL is validated
- Region-specific campaign is used
- Commercial configuration cannot change rankings

Epic E12: Operations and quality

Goal

Make the MVP supportable and trustworthy.

Stories

E12-S01: Import monitoring dashboard

E12-S02: Data-quality issue queue

E12-S03: Automated anomaly detection

E12-S04: Error-report form

E12-S05: Audit log

E12-S06: Backup and recovery procedure

E12-S07: Production alerting

34. Twelve-week sprint plan

Sprint 1: Foundation and schema

Weeks 1–2

Primary outcomes:

- Development environments
- CI pipeline
- Initial database
- Regions and currencies
- Product categories
- Canonical product schema

Tickets:

- E01-S01
- E01-S02
- E01-S03

- E02-S01
- E02-S02
- E03-S01
- E03-S02
- E03-S03

Sprint review demonstration:

- Create a region
 - Create a currency
 - Create a manufacturer
 - Create a DDR5 product
 - Run automated tests
-

Sprint 2: Retail sources and raw data

Weeks 3–4

Primary outcomes:

- Retailer configuration
- First import source
- Raw feed retention
- Raw listing and price records
- Exchange-rate import

Tickets:

- E02-S03
- E02-S04
- E04-S01
- E04-S02
- E04-S05
- E06-S01

Sprint review demonstration:

- Run an import
 - View import summary
 - Inspect retained source data
 - View raw price observations
 - View current exchange rate
-

Sprint 3: Matching and daily prices

Weeks 5–6

Primary outcomes:

- Automated matching
- Review queue
- Daily product-price calculations
- Corrections

Tickets:

- E05-S01
- E05-S02
- E05-S03
- E05-S04
- E06-S02
- E06-S03
- E06-S04
- E06-S05

Sprint review demonstration:

- Import unmatched listing
 - Review suggested product
 - Confirm match
 - Generate daily regional product price
 - Apply correction while preserving original data
-

Sprint 4: Indices and public charts

Weeks 7–8

Primary outcomes:

- Index definitions
- Eligibility rules
- Daily index values
- Quality rating
- Initial public category page

Tickets:

- E07-S01
- E07-S02
- E07-S03
- E07-S04
- E07-S05
- E07-S06
- E10-S04

Sprint review demonstration:

- Create a 32GB DDR5 index
 - Calculate 14 days of values
 - Show raw and indexed graph
 - Show quality status
 - Show product count and retailer count
-

Sprint 5: Events and forecasts

Weeks 9–10

Primary outcomes:

- Source register
- Event editor
- Regional impacts
- Forecast workflow
- Forecast page

Tickets:

- E08-S01
- E08-S02
- E08-S03
- E08-S04
- E08-S05
- E08-S06
- E09-S01
- E09-S02
- E09-S03
- E10-S06
- E10-S07

Sprint review demonstration:

- Create source
- Publish global event
- Add UK and Japan impacts
- Create forecast
- Submit for approval
- Publish forecast

Sprint 6: Public MVP and operational readiness

Weeks 11–12

Primary outcomes:

- Homepage
- Regional pages
- Product pages
- Affiliate links
- Monitoring
- Accessibility
- Private beta release

Tickets:

- E10-S01
- E10-S02
- E10-S03
- E10-S05
- E10-S08
- E10-S09
- E10-S10
- E11-S01
- E11-S02
- E11-S03
- E11-S04
- E11-S05
- E12-S01
- E12-S02
- E12-S03
- E12-S04
- E12-S05
- E12-S06
- E12-S07

Sprint review demonstration:

- Navigate complete public site
 - Select region
 - View category index
 - Read event and forecast
 - Follow disclosed affiliate link
 - View failed-import alert
 - Submit data-error report
-

35. Post-MVP backlog

Priority P1

- Newsletter integration
- Used-market index
- Forecast scorecard automation
- Regional preference cookie
- Chart CSV download
- Improved product search
- Retailer stock-history view
- Public correction log

Priority P2

- User accounts
- Watchlists
- Price alerts
- Complete-build cost index
- Canada and Australia
- French and Dutch retailer support
- Premium supporter tier

Priority P3

- GPU indices
 - CPU indices
 - Prebuilt PC tracking
 - Professional dashboard
 - External API
 - Webhook alerts
 - Commercial data exports
 - White-label charts
-

36. Definition of ready

A ticket is ready for development when:

- User or system value is clear.
 - Acceptance criteria are testable.
 - Dependencies are identified.
 - Required designs are attached.
 - Data-source assumptions are documented.
 - Security and privacy implications have been considered.
 - No unresolved product decision blocks implementation.
-

37. Definition of done

A ticket is complete when:

- Code is implemented.
 - Code has been reviewed.
 - Automated tests pass.
 - Acceptance criteria pass.
 - Error handling is included.
 - Logging is appropriate.
 - Documentation is updated.
 - Accessibility has been considered.
 - Security review is complete where relevant.
 - Feature works in staging.
 - Product owner or assigned reviewer accepts it.
 - No unresolved critical defect remains.
-

38. Release gates

Private beta gate

The system may enter private beta when:

- Four regions are configured.
- At least three regions have sufficient retail data.
- At least three category indices are public.
- Daily prices have run successfully for 30 days.
- Product-match accuracy exceeds 85%.
- Every index has a quality rating.
- Event and forecast workflows function.

- Affiliate disclosure is implemented.
- Privacy and cookie controls are live.
- Monitoring and backups are tested.
- Critical accessibility issues are resolved.

Public launch gate

The system may launch publicly when:

- Beta feedback has been reviewed.
 - No critical data-integrity issue remains.
 - At least ten meaningful events are published.
 - At least three active forecasts are published.
 - Methodology is complete.
 - Correction process has been tested.
 - Forecast-scorecard logic has been tested.
 - At least one complete forecast has been scored where timing permits.
 - Public pages pass performance and accessibility checks.
 - Affiliate links have been manually verified by region.
-

39. Initial team allocation

Product owner

Immediate tickets:

- Finalise launch index definitions
- Confirm region methodology
- Approve terminology
- Own backlog order
- Accept sprint output

Technical lead

Immediate tickets:

- Architecture
- Repository
- CI
- Database approach
- Security model
- Deployment

Data engineer

Immediate tickets:

- Source evaluation
- Import framework
- Canonical product catalogue
- Matching engine
- Daily calculations

Front-end developer

Immediate tickets:

- Design system
- Homepage
- Region selector
- Index charts
- Responsive layouts
- Accessibility

Market editor

Immediate tickets:

- Source taxonomy
- Event model
- Forecast template
- Launch content
- Methodology

Commercial lead

Immediate tickets:

- Affiliate programme register
- Regional retailer partnerships
- Disclosure requirements
- Revenue analytics
- Sponsor research

40. First team workshop agenda

The first implementation workshop should resolve the following.

Product decisions

- Final four launch regions
- Exact launch indices
- Minimum publication threshold
- Forecast terminology
- Recommendation terminology

Data decisions

- First retailer source
- Canonical product identifiers
- Tax-normalisation method
- Daily collection time
- Marketplace inclusion rules
- Outlier rules

Technical decisions

- Application framework
- Database
- Job scheduler
- Authentication
- Chart library
- Deployment method
- Monitoring platform

Editorial decisions

- Source tiers
- Publication approval
- Forecast approval
- Correction process
- Review cadence

Commercial decisions

- Affiliate programmes
- Link-redirection method
- Consent requirements
- Sponsor exclusions
- Commercial/editorial separation

Workshop output should be a signed-off decision log and a final Sprint 1 backlog.

41. Recommended first implementation milestone

The first meaningful vertical slice should be:

Import US DDR5 prices from one source, match them to canonical products, calculate a daily 32GB DDR5 index and display it on a private category page with its data-quality status.

This vertical slice proves:

- Source integration
- Product catalogue
- Product matching
- Raw price storage
- Daily derivation
- Index calculation
- API output
- Public charting
- Data-quality communication

The team should complete this before trying to build the full homepage or support every launch market.

42. Final MVP outcome

At the end of the initial delivery period, the platform should allow a visitor to:

1. Understand whether the global memory market is rising or falling.
2. See how that movement differs across supported regions.
3. View a regional price history in local currency.
4. Compare regional movement on a common indexed scale.
5. Inspect the market events associated with price movement.
6. Understand the evidence and uncertainty.
7. Read a buy-or-wait recommendation.
8. Review previous forecasts and their outcomes.
9. Follow a regionally appropriate affiliate link.
10. Verify how every public index and forecast was created.

Internally, the team should be able to:

1. Add new regions and retailers.
2. Import data safely.
3. Review product matches.
4. Correct derived results without deleting raw data.
5. Create and publish market events.
6. Publish accountable forecasts.
7. Monitor data quality.

8. Identify failed jobs and stale sources.
9. Review affiliate performance.
10. Expand categories without redesigning the entire system.